# Decision Trees as a Tool for Promoting Inclusion and Diversity in Adaptive Education

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***Abstract*—Inclusion in education is critical to achieving Sustainable Development Goal 4, which ensures equitable education for all. This study explores the application of decision tree algorithms to improve adaptive education, promoting inclusion and diversity in higher education. Decision trees were used to classify students into predefined categories based on various attributes such as gender, age, learning styles, and specific needs (for example, ADHD). The algorithm was trained on an experimental dataset and demonstrated high accuracy (95.94%) in classification. By dynamically adjusting educational content and accessibility options, this approach aims to provide personalized learning experiences tailored to individual needs. The findings highlight the potential of AI-driven adaptive learning systems to improve educational outcomes and foster an inclusive environment by accommodating various learning preferences and requirements. This research contributes to a greater understanding of how technology can be used to address educational inequities and suggests future directions to improve adaptive learning frameworks.**

***Index Terms*—adaptive education, Smart Learning Platforms, educational innovation, higher education, decision trees, inclusion and diversity.**

## I. Introduction

Inclusion in the context of education promotes the achievement of Sustainable Development Goal 4, "Quality of Education", which ensures inclusive and equitable quality education and promotes opportunities for lifelong learning for all [1]. From this stance, every effort to grant equitable access to education will allow learners to reach their full potential and develop opportunities to improve their livelihoods regardless of their physical, mental, socioeconomic, cultural, or linguistic abilities or contexts [2]. Inclusion also broadens the possibilities of understanding diversity as a powerful source of innovation, in which intersectionality, the combination of needs of traditionally marginalized or neglected populations [3], represents a myriad of opportunities to create different and more effective paths. Especially in the context of digital learning or e-learning, technological development opens a substantial number of possibilities regarding inclusion and accessibility. Inclusive education is not only about assisting people in need. It is about creating environments that promote and perform through the adaptability and adequacy of different types of responses according to what learners need. For example, adaptive evaluation [4] allows learners to achieve intermediate goals at their own pace while building the capacities and abilities needed to reach the main learning outcomes. Another good example of inclusive learning is the use of recommendation systems on e-learning platforms, which allow learners to follow personalized paths tailored to their interests and abilities [5]. Furthermore, Bardach et al. [6] note that automated feedback provides instant information on learners' performance, enhancing self-efficacy and classroom readiness, pointing out the relevant role of feedback and reflection during the learning process. All of these approaches are significantly improved when artificial intelligence (AI) is applied to determine the type of response, the adaptability paths, and, even more, the paths a learner should follow during a formative experience. Applying AI to adaptive e-learning has many benefits both for the learner and the administrator of the digital ecosystem, such as personalization, real-time monitoring and feedback, and process optimization. These capacities improve the learning experience [7] and promote inclusion and access to learners with neurodiversity. Furthermore, AI allows direct efforts to detect patterns and profiling strategies according to the diversity of learners [8] at each stage of the learning process, but is especially relevant in the early stages. In higher education, it becomes pertinent to broaden the possibilities for students to reach their highest potential by recognizing their demands toward a more effective and personalized education adapted to the diversity of students' learning styles and rhythms.

## II. Previous Work

Adaptive education uses advanced technologies and data-driven methodologies to design personalized learning experiences that address the diverse needs, preferences, and abilities of individual students, promoting equity and partic-

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ipation in educational environments. This approach seeks to enhance the educational process by providing personalized learning pathways that allow students to progress at their own pace. Adaptive education is particularly effective in fostering inclusion and diversity within higher education, as it can accommodate a wide range of learning requirements and preferences [9, 10]. The individualized approach benefits students and teachers, improving learning results and overall educational experiences. Integrating AI into adaptive education systems can dynamically adjust the difficulty level of tasks, recommend additional resources, and provide real-time feedback, optimizing the learning experience for every student [11, 12, 13]. In this sense, the VARK (Visual, Auditive, Reading, and Kinesthetic) model, which categorizes learning preferences into Visual, Auditory, Read/write, and Kinesthetic modalities [14], is relevant for adaptive education systems, as it underscores the importance of personalizing educational content to match individual learning styles [11, 15]. Studies have shown that the teaching-learning process can be more effective by recognizing and addressing these preferences, as students are more likely to engage with and understand the material presented in their preferred learning style [16, 17]. This approach ensures that all learners have the opportunity to succeed, thus contributing to a more equitable and effective education system.

## III. Method

This study adopted an experimental approach to evaluate the effectiveness of decision tree algorithms in promoting inclusion and diversity within adaptive education systems. The research design incorporated data collection, the development of an experimental model, and the validation of results in a real educational context.

The study involved a diverse sample of 244 higher education students from one of Mexico's largest private universities. Participants were selected to represent a variety of sociodemographic characteristics, including gender, age range, learning preferences according to the VARK model, and specific learning needs such as dyslexia or Attention Deficit and Hyperactivity Disorder (ADHD). Participation was voluntary and anonymous, ensuring confidentiality and ethical adherence in data handling.

Data collection was conducted through a survey designed to gather detailed information on each student's individual characteristics. The survey included questions addressing demographic variables such as gender, age range, and access to high-speed internet. It also captured information related to learning preferences and specific needs that might require accommodation. The responses collected formed a dataset that represents a diverse student population.

The research procedure began with the administration of the survey, allowing the collection of baseline data prior to the interaction of students with the adaptive learning platform. The data collected was processed and encoded into a structured tabular format suitable for algorithmic analysis. Using this dataset, a decision tree algorithm was trained to classify students into one of 14 predefined categories, which were designed based on key characteristics such as learning styles, internet access, and the presence of special needs. These classifications were then used to dynamically tailor educational content and accessibility options within the platform. The model training used 70% of the dataset, reserving 30% for testing and validation purposes. The results demonstrated a classification accuracy greater than 95%, confirming the reliability of the model for deployment in adaptive learning environments.

Data analysis involved quantitative techniques to assess the accuracy and effectiveness of the model. Metrics such as precision and sensitivity were applied to assess the performance of the algorithm. In addition, descriptive analyzes were performed to identify trends and patterns within student profiles, providing information on the diversity of the participant population.

One of the key strengths of this methodology lies in its ability to dynamically adapt learning materials and accessibility features to the specific needs of individual students. However, the study acknowledges certain limitations. The sample consisted predominantly of students aged 18 to 25, restricting the generalizability of the findings to broader or more diverse populations. In addition, the classification accuracy was heavily dependent on the completeness and precision of the input data, highlighting the need for robust data collection mechanisms to minimize potential misclassifications. These limitations underscore the importance of extending future research to include more heterogeneous samples and exploring complementary approaches that enhance data reliability and algorithmic robustness.

## IV. Results

### A. Entry Variables

To implement the classification algorithm, a set of foundational classes was defined, each designed to account for specific characteristics and values that distinguish them from each other [18]. The initial variables focused on demographic differentiators, which included students' gender and age range. Gender was categorized as "male", "female", "non-binary", and "prefer not to disclose". The age ranges were grouped into four categories: under 18, between 18 and 25, between 25 and 60, and above 60. The second set of variables was related to the environment and contextual background of the students. These variables included categorical indicators such as whether a student belonged to an ethnic group with indigenous Mexican roots and whether they had access to high-speed broadband internet. The final set of variables was primarily related to the students' learning capabilities. These included how students process information, based on the VARK learning style model, and variables identifying sensory limitations or specific learning needs that may require additional attention, such as ADHD.

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### B. Description of the Data Obtained

The data collection process involved the participation of 244 higher education students from Tecnologico de Monterrey. The students answered a poll that included their answers concerning the characteristics mentioned in the previous subsection. In the sample of students surveyed, the majority (98%) were found to be in the age range of 18 to 25 years. The remaining responses represented students younger than 18. For this study, data from students under 18 years of age were excluded from the data set. The gender identity of the students in the student sample was predominantly male, with 70% of the students polled. Meanwhile, 29% identified as female, and the remaining percent decided not to mention their gender. Regarding ethnicity, ninety-nine percent mentioned that they do not come from native roots, while only three students disclosed that they were from such lineages, representing Mixteco, Tlahuica, and Yaqui cultures.

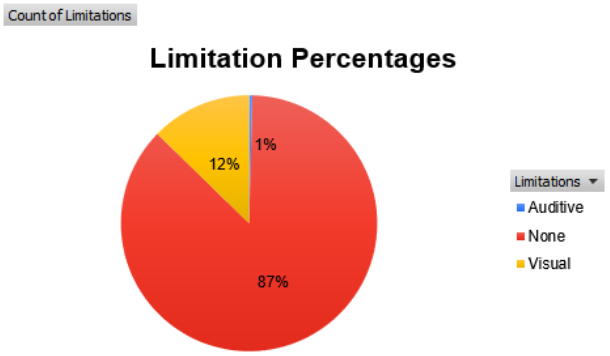



Fig. 1. Pie chart with the percentages for the represented communication and sensory limitations of students present in the study.

Figure 1. represents the distribution of students on communication and sensory limitations. The limitations most represented are visual and auditory, with 12% and 1%, respectively, while the remaining 87% do not appear to experience any of the listed limitations. According to the VARK model, most of the students identified with the visual learning style, representing 49%, as shown in Figure 2. The kinesthetic style was the second most common, with 31% of students, followed by the reading and auditory styles, which represented 9% and 7%, respectively.

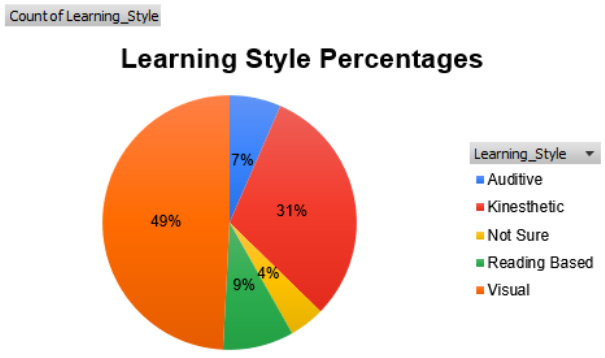



Fig. 2. Pie chart with the percentages for the represented learning styles of students present in the study.

When the study students were asked if they had access to high-speed broadband internet, the vast majority (98%) responded affirmatively. As shown in Figure 3, the most commonly reported special needs within the sample group were ADHD and dyslexia, at 11% and 2%, respectively. An additional 2% of the students reported having other special needs that were not specified in this study.

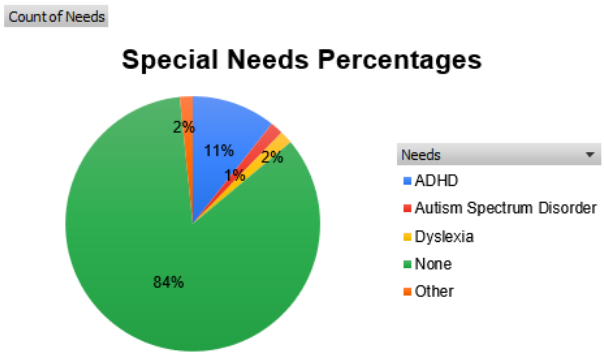



Fig. 3. Pie chart with the percentages for the represented special needs of students present in the study.

### C. Classification Algorithm

A decision tree machine learning algorithm was selected to be used as a classifier for this study. This algorithm was selected over other machine learning tools, such as neural networks, due to its principal use in tabular data while providing faster performance compared to similar tools [19]. This algorithm creates a tree-like graph trained on previously classified data that generates a structure of nodes and branches that the algorithm transits according to different values in a received input until it reaches a final node that provides the appropriate classification [20]. A working decision tree was built in Python, using the Sci-kit Learn machine learning library [21].

### D. Output Classes

Based on the survey variables applied to the students, a set of 14 output classifications was created for their different profiles. These classes consider key characteristics to create an adaptive content presentation and have a greater impact on the online experience and interaction between students and the e-learning platform.

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TABLE I
Summary of Output Classes

| Category | Class Type | Class Number | Description |
|---|---|---|---|
| Full Level | Visual | 1 | Visual learner with access to broadband internet |
| | Auditory | 2 | Auditory learner with access to broadband internet |
| | Kinesthesia | 3 | Kinesthetic learner with access to broadband internet |
| | Based on Reading | 4 | Reading learner with access to broadband internet |
| Low Level | Visual | 5 | Visual learner without access to broadband internet |
| | Auditory | 6 | Auditory learner without access to broadband internet |
| | Kinesthesia | 7 | Kinesthetic learner without access to broadband internet |
| | Based on Reading | 8 | Reading learner without access to broadband internet |
| Special Need | Dyslexia | 9 | Student with dyslexia |
| | Dysgraphia | 10 | Student with dysgraphia |
| | Dyscalculia | 11 | Student with dyscalculia |
| | Autism | 12 | Student with autism |
| | ADHD | 13 | Student with ADHD |
| Inconclusive | | 14 | The student didn't input enough data for classification. |

Table 1 shows how the 14 output classes are divided into four main categories. This classification was made considering the context from which students access the platform, whether students have conditions requiring special attention or input enough data to make an accurate classification. After a student registry falls under one of these categories, it is classified by either sensory profile or special need. The split between classes of different sensory profiles that can be seen in Table 1 as low level and full level considers, mainly, the connection speed context of a student. The low level is assigned to students without access to the broadband internet, and the full level is assigned to students who can access this resource. This division was implemented in this way considering the possibility that students with a slow internet connection may have less access to more dynamic and responsive online content. In this way, it is guaranteed that students can receive learning materials that can be delivered quickly and without problems based on the quality of their connection. Once defined, the classes assigned the best digital content that could be shown to them in a hierarchy. The hierarchy of content was made considering the possibility that specific topics in a course for the educational platform could be missed or not have a certain type of content made for it. The types of content available on the web learning platform were defined considering students' different characteristics, these contents include text, dynamic web content, videos, and questionnaires, forums, among other digital content designed to be dynamically adapted to their needs.

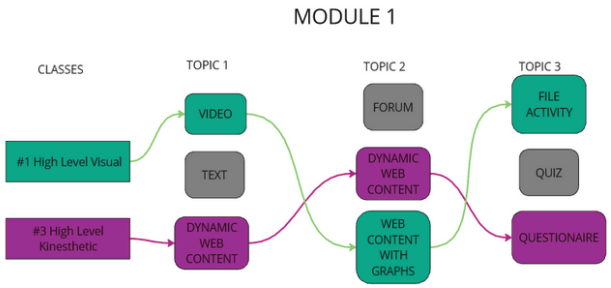



Fig. 4. Visualization of the dynamic assignment of contents for different classes.

Figure 4. shows a course module advances from a student of a certain class, the decision tree algorithm selects digital items that better fit their learning profile from a pool of pre-existing content. In addition to the dynamic selection of different types of content according to the profile of each student, customization in the presentation of content was implemented through the use of the UserWay tool. UserWay allows users to apply different functions related to accessibility [22]. It provides different accessibility options, which are implemented primarily by modifying the visual elements of web pages to meet the needs of certain individuals. Certain configurations of the Userway accessibility options were assigned to certain classes. The full-level and inconclusive classes did not have accessibility options. In contrast, all classes in the low-level category were assigned the options of disabled images and website animations, considering the possibility of more limited internet connections in this category. For students with special needs, the algorithm must make a more focused set of configurations based on each class. Figure 5 shows an example of the different accessibility options that the learning platform could automatically enable for students with different special needs profiles. These accessibility decisions, along with those made for other categories, were made to facilitate the experience of certain students and maintain their engagement with the platform.

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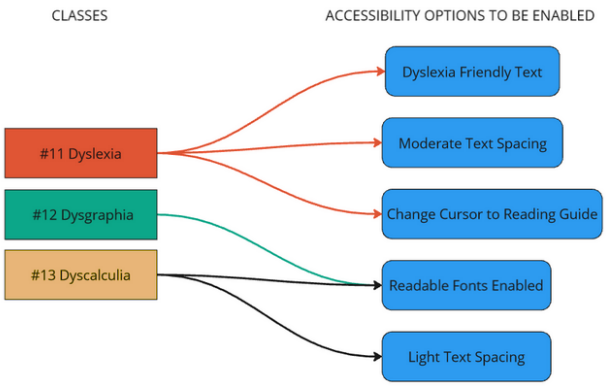



Fig. 5. Visualization of the dynamic enabling of accessibility options.

When using the gathered dataset containing all the responses given by the students pertaining to the variables mentioned in section 4.1, these values defined for the entry variables were converted into numerical values. When observing the assigned classes the group of students present in the study would mostly be high-level Visual and Kinesthetic learners with a representation of 43% and 25%, respectively, then followed by students with ADHD with 11%, high-level Reader students with 8%, high-level Auditory students with 5%, Inconclusive students with 4%, students in the Autism Spectrum with 2% and the remaining 2% split between Dyslexic and Low-Level Kinesthetic learners.

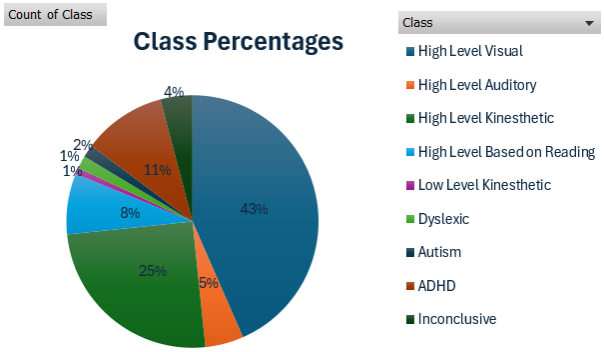



Fig. 6. Pie chart displaying the student classes represented in the study

A group of experts classified the data obtained from the responses of the students [18]. The input variables of each student with respect to their sociodemographic situation, together with their VARK learning style, allowed each student to be placed in one of the 14 output classes. Once the decision tree was trained in a proportion of 70% training data and 30% testing data, the algorithm's testing found that the algorithm had a precision of 95.94%.

## V. Discussion

The results of this study highlight the potential of decision tree algorithms to promote inclusive and adaptive learning in higher education contexts, as evidenced by the algorithm's high accuracy of 95.94%. This accuracy suggests that decision trees can effectively classify and customize learning experiences for diverse students when trained with well-defined student profiles. This aligns with the study's aim of promoting inclusion and diversity through adaptive technology. The use of decision tree algorithms in adaptive learning platforms significantly contributes to educational innovation's practical and theoretical dimensions. On a practical level, the ability of the decision tree to accommodate specific learning needs - such as ADHD, dyslexia, and autism, as well as different learning styles - demonstrates how educational content can be dynamically tailored to individual profiles. This adaptability is essential for creating inclusive learning environments that cater to neurodiverse populations and varying sensory needs, as evidenced by the student classifications shown in Figure 3. Theoretically, these findings support the extension of adaptive learning theories by integrating AI methods, further advancing personalised educational approaches in response to cognitive diversity. The successful classification of students into 14 categories based on accessibility and learning needs validates the algorithm's ability to distinguish between different profiles accurately. The incorporation of accessibility tools such as UserWay to adapt web content based on specific needs not only improves usability for students with neurodiversity but also supports the equitable engagement of diverse user groups. This alignment with user needs suggests that decision trees are effective for classification and highly applicable for real-time adaptive learning interventions in digital educational platforms. The limitations of this study are mainly related to the diversity of the sample. While the current dataset provides valuable insights, extending this model to more diverse student populations would increase the robustness of the results. In addition, the accuracy of the decision tree classifications is highly dependent on the completeness and accuracy of the student input data. Incomplete or inaccurate data submissions could lead to misclassifications, thereby compromising the quality of personalisation and potentially limiting the effectiveness of adaptive responses. Future iterations of this model should consider safeguards or alternative algorithms to minimise these risks.

## VI. Conclusions

This study explored implementing decision trees to foster inclusion and diversity within adaptive education systems. Decision trees were chosen over other machine learning methods due to their efficiency in processing structured data and their ability to provide rapid classification results. This approach contributes to the current understanding of adaptive education by offering a method that customizes learning experiences and optimizes the content delivery process based on individual learner profiles. While the decision tree algorithm demonstrated high accuracy, the study acknowledges potential limitations. The profiles of the students surveyed were mostly focused on only a few of the 14 classes created, so it will still be necessary to test the decision tree algorithm by applying the experiment to other sets of students with greater heterogeneity. Furthermore, the classification process relies heavily on the accuracy and completeness of input data from students. Incomplete or inaccurate data could lead to

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misclassification and subsequently affect the personalization of learning experiences. This study advances the current views on educational inclusion and diversity by demonstrating how AI-driven adaptive education systems can effectively cater to varied learning needs. The integration of accessibility tools such as UserWay further enhances the inclusivity of the platform, accommodating students with special needs by dynamically adjusting web content to meet their specific requirements. Implementing decision trees within adaptive education systems presents a promising approach to fostering inclusion and diversity. Future research can continue to enhance the effectiveness and accessibility of educational technologies, ultimately contributing to more equitable and personalized learning environments for all students. Future studies could extend the application of this decision tree model to other educational settings or more extensive, more diverse student populations to validate its generalisability. Additional research is recommended to explore how decision tree-based adaptive learning can be enhanced with real-time feedback mechanisms that can further refine content personalisation based on ongoing student performance. Integrating other machine learning models, such as neural networks in conjunction with decision trees, may offer improved accuracy and adaptability in adaptive learning systems, promoting more accurate, responsive, and inclusive educational environments.

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